
Infrared Spectroscopy of Atmospheric Trace Gases (Greenhouse Effect)

Advanced Lab course
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Date of Experiment : January 29, 2026

Date of Submission : March 3, 2026

Contents

List of Figures	ii
List of Tables	ii
1 Introduction	1
2 Theoretical Background	1
2.1 Infrared Region	1
2.2 Molecular Degrees of Freedom	1
2.3 Vibrational Energy of a Diatomic Molecule	2
2.4 Rotational Energy of a Diatomic Molecule	2
2.5 Rovibrational Transitions	3
2.6 Intensity Distribution and $J_{\max}$	4
2.7 Fourier-Transform Spectrometer	5
2.8 Lambert–Beer Absorption Law	7
2.9 Barometric Altitude Formula	8
3 Experimental Methods	10
3.1 Apparatus	10
3.2 Measurement Principle	10
3.3 Procedure	10
4 Evaluation of the measured spectra	11
4.1 Description of the spectra	11
4.2 Blackbody Temperature	12
4.3 Identification of the molecules	12

List of Figures

1	Schematic setup of a Michelson interferometer used in an FTIR spectrometer.	6
2	Examples of spectra (left) and the corresponding interferograms (right) . . .	7
3	Sun Spectra	11
4	Molecular Regions Highlighted	12
5	Absorption Band	13

List of Tables

1	Observed and Literature Band Origins of Atmospheric Trace Gases	13
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1 Introduction

The Earth's radiative balance is strongly influenced by the absorption of infrared (IR) radiation by atmospheric trace gases. While the atmosphere is largely transparent to incoming solar radiation in the visible range, several molecular species absorb outgoing terrestrial radiation in the infrared spectral region. This absorption process forms the physical basis of the greenhouse effect.

Molecules interact with electromagnetic radiation when the radiation frequency matches allowed transitions between quantized molecular energy levels.

In the infrared region, these transitions correspond primarily to vibrational and rotational motions of molecules. Only molecules whose vibration leads to a change in electric dipole moment are IR active.

Greenhouse gases such as CO₂, H₂O, and CH₄ possess vibrational modes that produce dipole moment changes and therefore absorb infrared radiation efficiently. In contrast, homonuclear diatomic molecules such as N₂ and O₂ do not exhibit dipole moment changes during vibration and are therefore largely IR inactive.

In this experiment, infrared absorption spectra of atmospheric gases were measured using a Fourier Transform Infrared (FTIR) spectrometer. The aim was to identify characteristic absorption bands of greenhouse gases and relate these spectral features to molecular structure and the physical mechanism of the greenhouse effect.

2 Theoretical Background

2.1 Infrared Region

The infrared region spans approximately

$$\lambda \approx 0.7 \mu\text{m to } 1000 \mu\text{m.} \quad (1)$$

In spectroscopy, the wavenumber $\tilde{\nu}$ is commonly used:

$$\tilde{\nu} = \frac{1}{\lambda}. \quad (2)$$

Molecular vibrational transitions typically occur in the mid-infrared region:

$$4000\text{--}400 \text{ cm}^{-1}. \quad (3)$$

2.2 Molecular Degrees of Freedom

For a molecule consisting of N atoms:

$$\text{Total degrees of freedom} = 3N \quad (4)$$

$$\text{Translational degrees} = 3 \quad (5)$$

Rotational degrees of freedom:

- Linear molecule: 2
- Nonlinear molecule: 3

Vibrational degrees of freedom:

- Linear molecule: $3N - 5$
- Nonlinear molecule: $3N - 6$

Examples:

CO_2 (linear, $N = 3$):

$$3N - 5 = 4 \text{ vibrational modes} \quad (6)$$

H_2O (nonlinear, $N = 3$):

$$3N - 6 = 3 \text{ vibrational modes} \quad (7)$$

2.3 Vibrational Energy of a Diatomic Molecule

In the harmonic oscillator approximation:

$$E_v = \hbar\omega \left(v + \frac{1}{2} \right), \quad (8)$$

where $v = 0, 1, 2, \dots$

Infrared transitions mainly follow the selection rule:

$$\Delta v = \pm 1. \quad (9)$$

2.4 Rotational Energy of a Diatomic Molecule

For a rigid rotor:

$$E_J = BJ(J + 1), \quad (10)$$

with

$$B = \frac{\hbar^2}{2I}, \quad (11)$$

where $J = 0, 1, 2, \dots$ and I is the moment of inertia.

Selection rule:

$$\Delta J = \pm 1. \quad (12)$$

The combination of vibrational and rotational transitions leads to rovibrational bands consisting of P- and R-branches.

2.5 Rovibrational Transitions

To obtain a spectrum from rotational and vibrational energy levels, selection rules must be determined that specify which transitions are allowed. These rules follow from the transition dipole moment matrix elements

$$|\langle J'M'|\hat{\mu}_i|JM\rangle|^2 = \left| \int \psi_{J'M'}^* \hat{\mu}_i \psi_{JM} d^3r \right|^2. \quad (13)$$

A transition is allowed only if this integral is nonzero. Whether it vanishes or not is determined by the symmetry properties of the molecular wavefunctions.

For a diatomic molecule in the harmonic oscillator and rigid rotor approximation, the selection rules are:

Vibrational

$$\Delta v = \pm 1 \quad (14)$$

Rotational

$$\Delta J = \pm 1 \quad (15)$$

Since vibrational energy spacings are much larger than rotational energy spacings ($E_{\text{vib}} \gg E_{\text{rot}}$), infrared radiation excites vibrational and rotational degrees of freedom simultaneously. Pure vibrational transitions without rotational contribution are therefore not observed.

Energy Levels

Restricting the consideration to transitions from the vibrational ground state ($v = 0$) to the first excited state ($v = 1$), the rovibrational energy levels are given by

$$E(v, J) = h\nu_0 \left(v + \frac{1}{2} \right) + BJ(J+1), \quad (16)$$

where ν_0 is the fundamental vibrational frequency and

$$B = \frac{\hbar^2}{2I} \quad (17)$$

is the rotational constant.

Transition Energies

The energy difference for a rovibrational transition becomes

$$\Delta E = h\nu_0 \pm 2BJ. \quad (18)$$

Thus, transitions with $\Delta J = +1$ or $\Delta J = -1$ produce lines with positive or negative frequency shifts relative to the pure vibrational frequency ν_0 .

Band Structure

The observed spectrum, known as a rovibrational band, is divided into:

- P-branch ($\Delta J = -1$)
- R-branch ($\Delta J = +1$)

Because transitions with $\Delta J = 0$ are forbidden, no spectral line appears exactly at ν_0 . The spacing between neighboring lines is

$$\nu(J+1) - \nu(J) = 2B, \quad (19)$$

which shows that all lines within a branch are equally spaced.

2.6 Intensity Distribution and $J_{\max}$

The intensity distribution within a rovibrational absorption band of a diatomic molecule is determined by the thermal population of rotational states and their degeneracy.

At thermal equilibrium, the population of a rotational level J follows Boltzmann statistics:

$$N_J \propto (2J+1) \exp\left(-\frac{BJ(J+1)}{k_B T}\right). \quad (20)$$

The factor $(2J+1)$ accounts for the rotational degeneracy due to the possible orientations of the angular momentum.

For small J , the degeneracy term dominates and the population increases with J . For larger J , the exponential Boltzmann factor becomes dominant and the population decreases again.

Therefore, the line intensities within the P- and R-branches increase up to a maximum and then decrease.

To determine the position of the intensity maximum, $J_{\max}$, J is treated as a continuous variable and the derivative of the population function is set to zero:

$$\frac{d}{dJ} \left[(2J+1) \exp\left(-\frac{BJ(J+1)}{k_B T}\right) \right] = 0. \quad (21)$$

This leads to the approximate expression

$$J_{\max} \approx \sqrt{\frac{k_B T}{2B}} - \frac{1}{2}. \quad (22)$$

Hence, $J_{\max}$ increases with temperature and decreases with increasing rotational constant B .

2.7 Fourier-Transform Spectrometer

A Fourier-Transform Spectrometer (FTIR) is used to measure infrared absorption spectra with high spectral resolution and high signal-to-noise ratio. Instead of dispersing light into individual frequencies, the FTIR records an interferogram in the optical path difference domain and converts it into a spectrum by Fourier transformation.

Principle of Operation

The central component of an FTIR spectrometer is a Michelson interferometer consisting of:

- Infrared source
- Beam splitter
- Fixed mirror
- Movable mirror
- Detector

The beam splitter divides the incoming broadband infrared radiation into two beams. One beam is reflected toward a fixed mirror, while the other is transmitted toward a movable mirror. After reflection, the two beams return to the beam splitter, where they recombine and interfere with each other.

The movable mirror changes position during the measurement, which varies the optical path difference between the two beams in a controlled manner. This variation produces constructive and destructive interference depending on the wavelength and path difference. As a result, the detected intensity changes as a function of mirror displacement, generating an interference signal called an interferogram. This interferogram contains information about all wavelengths present in the source simultaneously. By applying a Fourier transform to the interferogram, the spectral intensity of the infrared radiation can be obtained as a function of wavenumber.



Figure 1: Schematic setup of a Michelson interferometer used in an FTIR spectrometer.

For a given wavelength, the detected intensity varies with mirror position due to constructive and destructive interference. The positions of maximum and minimum intensity depend on the wavelength, since the phase difference between the interfering beams is determined by the optical path difference.

Because the interferometer does not separate wavelengths, the detector measures only the total intensity, which is the sum of the contributions from all wavelengths present. Thus, the measured signal represents the superposition of all spectral components.

If the light source emits n discrete wavelengths, measuring the intensity at n different mirror positions provides a system of n equations that can be solved to determine the spectral intensities. Increasing the maximum mirror displacement improves the spectral resolution and allows closely spaced wavelengths to be resolved.



Figure 2: Examples of spectra (left) and the corresponding interferograms (right)

2.8 Lambert–Beer Absorption Law

According to the Lambert–Beer absorption law, the transmitted intensity at frequency ν after passing through an absorbing medium is given by

$$I(\nu) = I_0(\nu) \exp[-k(\nu) p l], \quad (23)$$

where

- $I_0(\nu)$ is the incident intensity,
- $I(\nu)$ is the transmitted intensity,
- $k(\nu)$ is the absorption coefficient [$\text{cm}^{-1}\text{atm}^{-1}$],
- p is the gas pressure [atm],
- l is the absorption path length [cm].

Column Density Form

Alternatively, the Lambert–Beer law can be expressed in terms of the column density N :

$$I(\nu) = I_0(\nu) \exp[-\beta(\nu) N], \quad (24)$$

where

- $\beta(\nu)$ is the molecular absorption cross section [cm^2],
- N is the column density, i.e. the total number of absorbing particles per unit area [cm^{-2}].

Absorption Line Strength

The strength of an absorption line or an entire absorption band is obtained by integrating the absorption coefficient over frequency:

$$S = \int k(\nu) d\nu. \quad (25)$$

The integrated line strength S is an intrinsic molecular property and is independent of pressure and path length.

2.9 Barometric Altitude Formula

The barometric altitude formula describes the decrease of atmospheric pressure with increasing altitude under the assumption of hydrostatic equilibrium and constant temperature.

The pressure at height h is given by

$$p(h) = p(h_0) \exp\left(-\frac{gM(h-h_0)}{RT}\right), \quad (26)$$

where

- $p(h)$ is the pressure at altitude h ,
- $p(h_0)$ is the pressure at reference altitude h_0 (e.g. sea level),
- $g = 9.81 \text{ m s}^{-2}$ is the gravitational acceleration,
- M is the molar mass of the gas,
- $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ is the universal gas constant,
- T is the absolute temperature,
- $(h - h_0) = \Delta h$ is the altitude difference.

Particle Density

Using the ideal gas law,

$$p = nk_B T, \quad (27)$$

the particle number density $n(h)$ becomes

$$n(h) = \frac{p(h)}{k_B T(h)}. \quad (28)$$

The total number of particles along the atmospheric column is obtained by integrating the particle density over height:

$$N = \int_0^{h_{\max}} n(h) dh. \quad (29)$$

Temperature Profile in the Troposphere

In the troposphere (up to approximately 10 km), the temperature decreases approximately linearly with altitude. The average lapse rate is

$$\Gamma \approx 6 \text{ K km}^{-1}. \quad (30)$$

The temperature at altitude h can therefore be approximated by

$$T(h) = T(h_0) - \Gamma(h - h_0), \quad (31)$$

where

- $T(h)$ is the temperature at altitude h ,
- $T(h_0)$ is the temperature at sea level,
- Γ is the temperature lapse rate.

Above the troposphere, the particle density decreases significantly and the temperature profile deviates from this linear approximation.

3 Experimental Methods

3.1 Apparatus

The measurements were performed using a Fourier Transform Infrared (FTIR) spectrometer consisting of:

- Broadband infrared source
- Michelson interferometer
- Beam splitter
- Fixed mirror
- Movable mirror
- Gas cell / atmospheric path
- Infrared detector
- Computer for Fourier transformation

3.2 Measurement Principle

The Michelson interferometer splits the incident radiation into two beams. One beam is reflected by a fixed mirror, the other by a movable mirror. The recombined beams interfere depending on the optical path difference. The recorded interferogram is Fourier transformed to obtain the spectrum.

3.3 Procedure

1. The spectrometer was aligned and initialized.
2. A background spectrum was recorded.
3. Atmospheric air was measured under laboratory conditions.
4. Spectra were recorded in the mid-infrared region.
5. Interferograms were Fourier transformed by the instrument software.
6. Absorbance spectra were obtained by comparison with the background measurement.

4 Evaluation of the measured spectra

After performing the experiment, we obtained the spectra of N_2O and lab air. Due to the unclear weather we were not able to obtain the spectrum of the sun. But the required sun spectra were then handed to us. The following section will evaluate the spectra.

4.1 Description of the spectra

The absorption spectrum of sun is typical for infrared (IR) region spectrum with distinct absorption bands of CO_2 and H_2O . In the spectrum of lab air we also see the absorption bands of CO_2 and H_2O . But they are much less pronounced because the path length of the light through atmosphere is much higher than the path length of the light through lab air.

The solar spectra have the maximum at the wavenumber of 2615.67cm^{-1} . This is not consistent with the expected blackbody radiation maximum. That is because we are measuring the spectrum in the infrared. This is out of the maximum of the sun's peak radiation output. The second reason is the atmosphere of the earth. Earth absorbs lots of radiation and hence shifts the maximum to unrealistic values. This will further complicate the calculation of the true maximum of the sun's spectrum.

In addition to that the sensitivity of the instrument is different for different wavenumbers. Which also adds to the shift of the maximum.

The spectrum is also dense with absorption bands. This is caused by typical IR absorbing molecules. This can be seen in the figure 3.

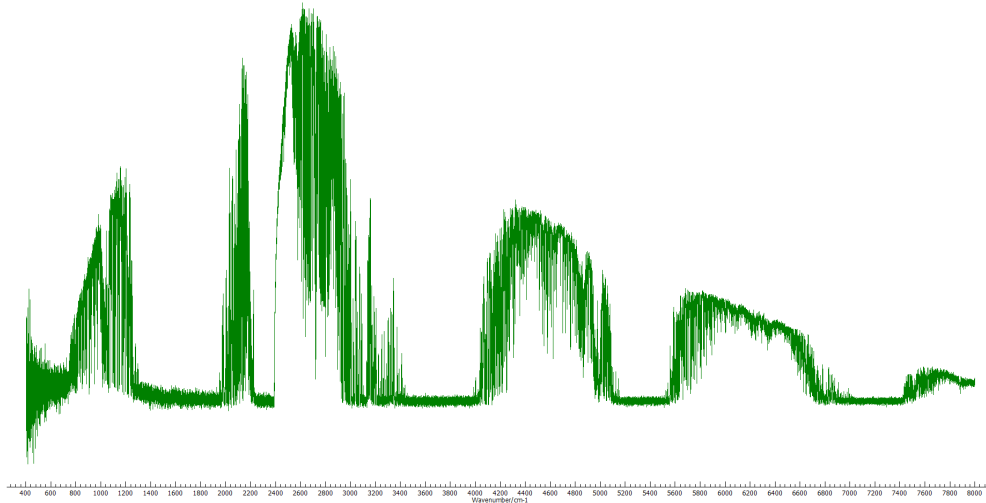


Figure 3: Absorption spectra of the sun with dense regions of absorption bands.

The spectrum of sun is very wide compared to the other two, spanning from 300 cm^{-1} to 8000 cm^{-1} . This gives us limited comparison of the spectra. We managed to identify 3 most prominent absorption bands. In the figure 4 we have identified the corresponding molecules responsible for the absorption bands. The most prominent is water. Because of its V-shaped structure it has many vibrational modes. Which are $3N - 6$ for non-linear molecules. Because of 3 fundamental vibrations, H_2O has no clear P and R branches compared to CO_2 . Which

is second molecule we found. The visible P and R branches are result of the linear structure of CO_2 .

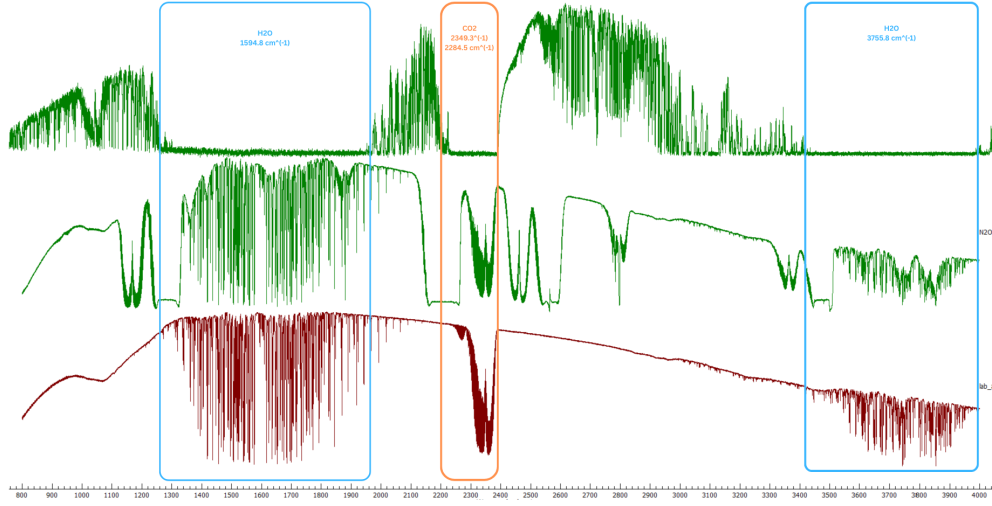


Figure 4: This is the comparison of the spectra of sun(top), N_2O , and lab air , showing absorption bands.

4.2 Blackbody Temperature

With the help of intensity maximum of the sun spectra we can calculate the temperature of the sun. Which is given by the Wien's displacement law:

$$\lambda_{\max} = \frac{b}{T} \quad (32)$$

with $b = 0.29cm \cdot K$.

With the maximum of the spectrum at $2615.67cm^{-1}$, the temperature of the sun came out to be $758.54K$. This is much lower than actual temperature of the sun which is around $5772K$. This is because of the reasons mentioned in the previous section 4.1.

In addition to that we also calculated the temperature of the lab air and N_2O using the same method. The temperatures came out to be $527.69K$ and $415.2K$. This again is lower than the actual temperature of the laser used for the experiment which is around $1000K$. For this the reason is most likely the sensitivity of the instrument.

4.3 Identification of the molecules

By closely examining the solar spectra we found lot of absorption bands with P and R branches. The results are listed in a table 1

Apart from the usual candidates we also found the absorption band near $1044.3cm^{-1}$. This band was too hard to ignore with a prominent P and R branch figure:5 As for what molecule this band is from, We are not sure. Only molecule with strong band at this wavenumber is ozone. But ozone is not linear and hence should not have such clear P and

Observed Band Origin (cm^{-1})	Literature Value (cm^{-1})	Identified Molecule
2462.6	2461.5	N_2O
2563.3	2563.5	N_2O
4855.5	4860.5	CO_2
4977.8	4983.5	CO_2
6228.4	6231.0	CO_2
6345.8	6351.0	CO_2

Table 1: Observed and Literature Band Origins of Atmospheric Trace Gases

R branches. In literature we see the band of ozone at 1042cm^{-1} [1]. This is close to our observation but we are not sure.

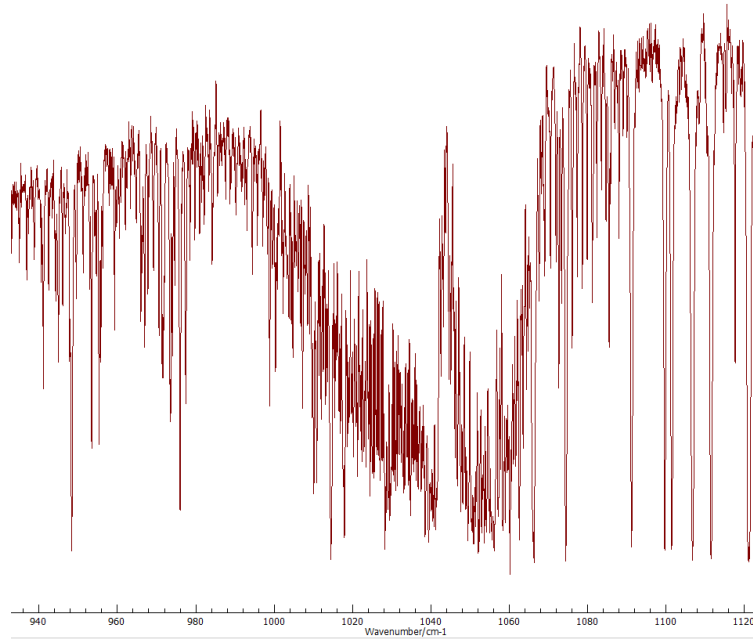


Figure 5: Absorption band of with P and R branches.

References

- [1] V. Gorshelev, A. Serdyuchenko, M. Weber, W. Chehade, and J. P. Burrows. High spectral resolution ozone absorption cross-sections – part 1: Measurements, data analysis and comparison with previous measurements around 293 k. *Atmospheric Measurement Techniques*, 7(2):609–624, 2014.